

Experiments on Lowest-Confidence Words in GPT-3.5-Turbo and GPT-4-Turbo

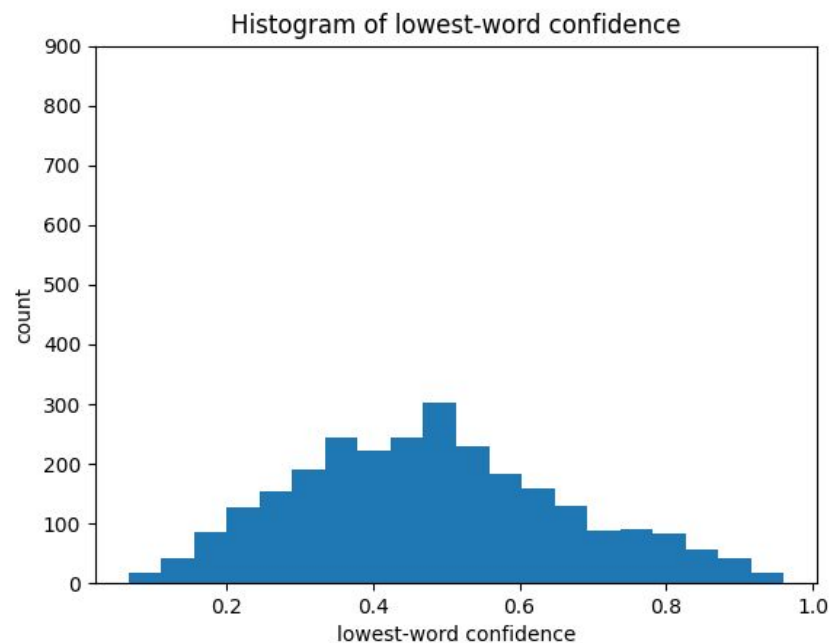
Maryam Naderi

Here is the example of prompt for next experiment

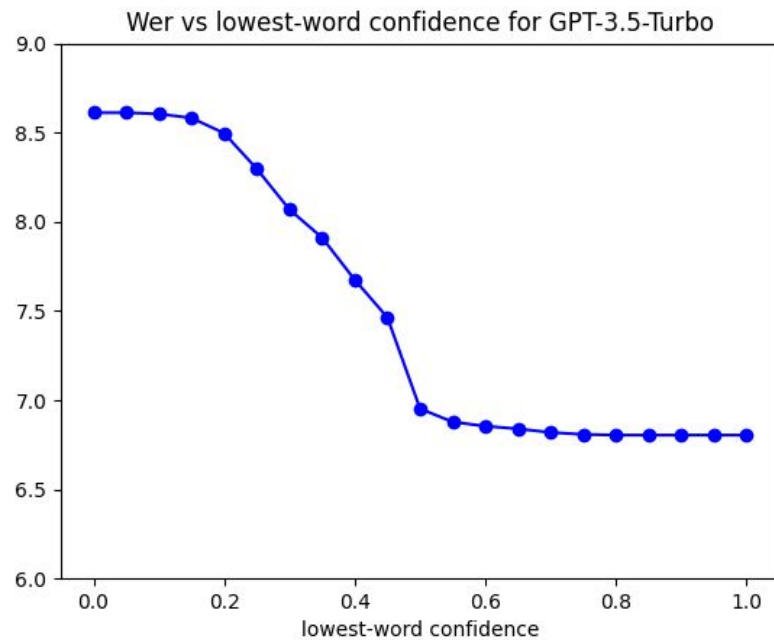
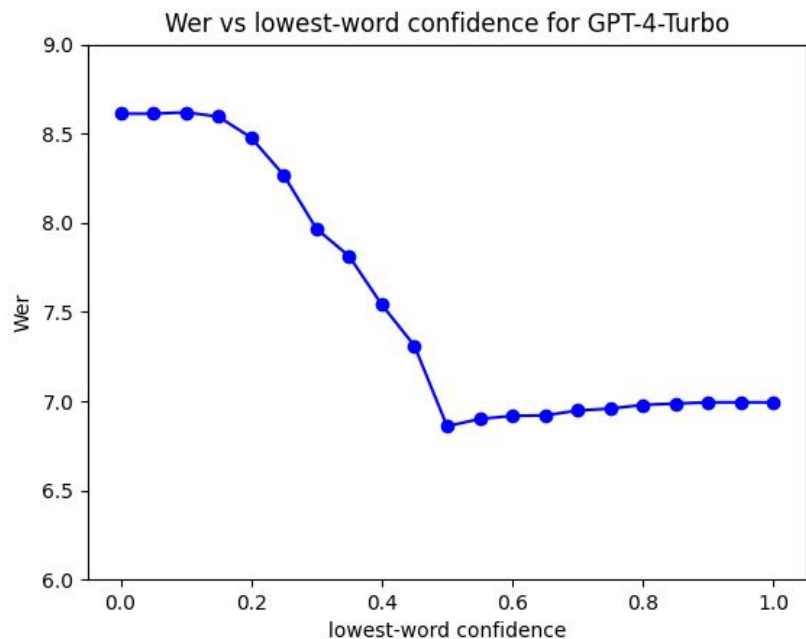
```
{  
    'role': 'system',  
    'content': f"""You are a helpful assistant that corrects ASR errors. \  
You will be presented with an ASR transcription delimited by {delimiter} characters,\  
and your task is to rectify any errors in it. \  
Provide the most probable corrected transcription in string format. \  
Do not change the case, for example, lower case or upper case, in the transcription. \  
Do not output any additional text that is not the corrected transcription. \  
Do not write any explanatory text that is not the corrected transcription.  
"""  
}
```

We obtained the results, along with their associated lowest word-level confidence scores, from the “exp_without_lowest_word_confidence” for both GPT-3.5-Turbo and GPT-4-Turbo.

Histogram plot of the lowest-confidence words of the Librispeech transcriptions



WER vs lowest-confidence plots for GPT-3.5-Turbo and GPT-4-Turbo



My team's suggestions about this experiment

Mathew and Enno suggested me to try different whisper models (small-medium-large) for both gpt-3.5-turbo-1106 and gpt-4-1106-preview and compare all the results of these model together.

Mathew also suggested me for next step we can use LF-MMI or Hybrid-HMM/ANN instead of whisper model.

Thank you